Homework 4 Solution

Problem 1.

(a) True. If the primal has m functional constraints and n variables, then the dual has n functional constraints and m variables. Thus both sums are $m + n$.

(b) False. Before optimality, the simplex method does not necessarily have a dual feasible corner-point solution with the same objective value. If both primal and dual feasible solutions had equal objective values, weak duality would imply that both were already optimal.

(c) False. If the primal problem is unbounded, then the dual problem must be infeasible by weak duality. Hence the dual problem has no optimal value.

Problem 2.

(a) The dual problem is

$$\begin{aligned} \text{Minimize} \quad & W = y, \\ \text{subject to} \quad & y \geq 2, \\ & -y \geq -4, \\ & y \geq 0. \end{aligned}$$

Equivalently, $2 \leq y \leq 4$, so the dual optimal solution is $y^* = 2$ and $W^* = 2$.

(b) Since $y^* > 0$, complementary slackness implies that the primal constraint is tight:

$$x_1 - x_2 = 1.$$

The second dual constraint has positive slack at $y^* = 2$, because $-y^* = -2 > -4$, so complementary slackness gives $x_2^* = 0$. Therefore $x_1^* = 1$. The primal optimal solution is

$$x^* = (1, 0), \qquad Z^* = 2.$$

(c) If the coefficient of x_1 is c_1, the dual constraints become

$$y \geq c_1, \qquad y \leq 4, \qquad y \geq 0.$$

The dual problem is infeasible exactly when $c_1 > 4$. For these values, the primal problem is feasible and has an unbounded objective function.

Problem 3.

(a) The complementary slackness conditions are

$$y_1(5x_1 + x_2 + 3x_3 - 8) = 0, \qquad y_2(3x_1 + 2x_2 - 3) = 0,$$

$$x_1(13 - 5y_1 - 3y_2) = 0, \qquad x_2(10 - y_1 - 2y_2) = 0, \qquad x_3(6 - 3y_1) = 0.$$

(b) Since $x_1 = 1 > 0$ and $x_3 = 1 > 0$, the first and third dual constraints must be tight:

$$5y_1 + 3y_2 = 13, \qquad 3y_1 = 6.$$

Thus $y_1 = 2$ and $y_2 = 1$. This solution is dual feasible because it satisfies all dual constraints:

$$y_1 + 2y_2 = 4 \le 10.$$

The primal and dual objective values are both 19, so $(y_1, y_2) = (2, 1)$ is dual optimal.

Problem 4.
Here

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \pi = \begin{bmatrix} \frac{9}{20} \\ \frac{7}{20} \\ \frac{3}{10} \\ \frac{13}{20} \end{bmatrix}, \qquad q = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 1 \end{bmatrix}.$$

(a) The primal problem is

$$\begin{array}{ll} \text{Maximize} & \frac{9}{20}x_1 + \frac{7}{20}x_2 + \frac{3}{10}x_3 + \frac{13}{20}x_4 - w \\ \text{subject to} & x_1 + x_4 - w \le 0, \\ & x_2 + x_4 - w \le 0, \\ & x_3 - w \le 0, \\ & 0 \le x_1 \le 2, \quad 0 \le x_2 \le 2, \quad 0 \le x_3 \le 2, \quad 0 \le x_4 \le 1. \end{array}$$

The variable w is unrestricted in sign.

(b) Let $p_R, p_B, p_G \ge 0$ correspond to the three worst-case payoff constraints, and let $y_j \ge 0$ correspond to the upper bound $x_j \le q_j$. The dual is

$$\begin{array}{ll} \text{Minimize} & 2y_1 + 2y_2 + 2y_3 + y_4 \\ \text{subject to} & p_R + y_1 \ge \frac{9}{20}, \\ & p_B + y_2 \ge \frac{7}{20}, \\ & p_G + y_3 \ge \frac{3}{10}, \\ & p_R + p_B + y_4 \ge \frac{13}{20}, \\ & p_R + p_B + p_G = 1, \\ & p_R, p_B, p_G, y_1, y_2, y_3, y_4 \ge 0. \end{array}$$

Here p_R, p_B, p_G are state prices, or risk-neutral probabilities, for the three tournament outcomes. The variables y_j are scarcity values for the quantity caps q_j.

(c) Under strict complementarity, the complementary slackness conditions imply the following acceptance rules for each order j:

Acceptance of order j	Condition
$x_j = q_j$	$a_j^T p < \pi_j$ and $y_j = \pi_j - a_j^T p > 0$
$0 < x_j < q_j$	$a_j^T p = \pi_j$ and $y_j = 0$
$x_j = 0$	$a_j^T p > \pi_j$ and $y_j = 0$

Here $a_j^T p$ is the state-price value of one unit of order j. Thus high-profit orders are fully accepted, exactly fair orders are partially accepted, and underpriced orders from the bookmaker's perspective are not accepted.

(d) One optimal primal solution is

$$x^* = (2, 2, 2, 0), \qquad w^* = 2,$$

with objective value

$$\frac{9}{20} \cdot 2 + \frac{7}{20} \cdot 2 + \frac{3}{10} \cdot 2 + \frac{13}{20} \cdot 0 - 2 = \frac{1}{5}.$$

One optimal dual solution is

$$p_R^* = \frac{17}{40}, \qquad p_B^* = \frac{13}{40}, \qquad p_G^* = \frac{1}{4}, \qquad y^* = \left(\frac{1}{40}, \frac{1}{40}, \frac{1}{20}, 0\right).$$

It is feasible since

$$\frac{17}{40} + \frac{1}{40} = \frac{9}{20}, \qquad \frac{13}{40} + \frac{1}{40} = \frac{7}{20}, \qquad \frac{1}{4} + \frac{1}{20} = \frac{3}{10}, \qquad \frac{17}{40} + \frac{13}{40} + 0 = \frac{3}{4} \geq \frac{13}{20}.$$

The dual objective value is

$$2 \cdot \frac{1}{40} + 2 \cdot \frac{1}{40} + 2 \cdot \frac{1}{20} + 0 = \frac{1}{5},$$

so the two solutions are optimal by strong duality.